1 SU(2) as a Double Cover of SO(3)

Step 1: Parametrising rotations Each rotation in $\mathbb{R}^3$ is determined by an oriented axis $v \in S^2$ and an angle $\varphi \in [0, \pi]$. The only identifications are

$$(v, \pi) \sim (-v, \pi), \qquad (v, 0) \sim (w, 0) \quad \text{for all } v, w \in S^2.$$

Thus, if we identify $\mathbb{R}P^3$ with the closed 3-ball of radius π whose antipodal boundary points are identified, namely

$$v \cdot \pi \sim -v \cdot \pi \qquad \text{for all } v \in S^2,$$

then the map

$$\mathrm{SO}(3) \longrightarrow \mathbb{R}P^3, \qquad (v, \varphi) \longmapsto [\, v \cdot \varphi \,]$$

is well defined. Here $[\, v \cdot \varphi \,]$ denotes the corresponding point in the quotient space. The rotation angle φ, taken modulo 2π, depends continuously on the rotation, and the axis v also depends continuously on the rotation whenever $\varphi \neq 0$.

Step 2: SU(2) $\cong S^3$ The group $\mathrm{SU}(2)$ consists of 2×2 unitary matrices of determinant 1:

$$U = \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix}, \qquad |\alpha|^2 + |\beta|^2 = 1, \quad \alpha, \beta \in \mathbb{C}.$$

Writing $\alpha = a_0 + ia_3$, $\beta = a_2 + ia_1$ the constraint becomes $a_0^2 + a_1^2 + a_2^2 + a_3^2 = 1$, so $\mathrm{SU}(2) \cong S^3$ (the unit 3-sphere in $\mathbb{R}^4$).

Step 3: The covering map Identify $\mathbb{R}^3$ with the space of traceless skew-Hermitian matrices (or equivalently, with purely imaginary quaternions)

$$\mathbf{x} = x_1\mathbf{i} + x_2\mathbf{j} + x_3\mathbf{k} \;\longleftrightarrow\; X = \begin{pmatrix} ix_3 & ix_1 + x_2 \\ ix_1 - x_2 & -ix_3 \end{pmatrix}.$$

For each $U \in \mathrm{SU}(2)$, the map

$$\Phi(U) : \; X \;\longmapsto\; UXU^{-1}$$

is a **rotation** of $\mathbb{R}^3$ (it preserves the inner product $\langle X, Y \rangle = -\frac{1}{2}\mathrm{tr}(XY)$). This defines a smooth group homomorphism

$$\Phi : \mathrm{SU}(2) \;\longrightarrow\; \mathrm{SO}(3).$$

Step 4: The kernel is $\mathbb{Z}/2$ The kernel of Φ consists of those U that commute with every X, which forces $U = \pm I_2$. Hence

$$\ker \Phi = \{+I_2, \, -I_2\} \cong \mathbb{Z}/2\mathbb{Z}.$$

In terms of the parametrisation above, the two matrices

$$U(\varphi, v) = \cos\tfrac{\varphi}{2}\, I_2 + \sin\tfrac{\varphi}{2} \begin{pmatrix} iv_3 & iv_1 + v_2 \\ iv_1 - v_2 & -iv_3 \end{pmatrix}$$

and $-U(\varphi, v) = U(\varphi + 2\pi, v)$ both map to the *same* rotation (v, φ). This is the origin of the double cover: going around a 2π loop in SO(3) lifts to a path from U to $-U$ in SU(2); you need a 4π rotation to return to the identity in SU(2).

Step 5: Conclusion Since Φ is surjective and has kernel $\mathbb{Z}/2$, the first isomorphism theorem gives

$$\boxed{\mathrm{SO}(3) \;\cong\; \mathrm{SU}(2)/\{\pm I_2\} \;\cong\; S^3/\mathbb{Z}_2 \;\cong\; \mathbb{RP}^3.}$$

In other words, $\mathrm{SU}(2) \cong S^3$ is the **universal (double) cover** of $\mathrm{SO}(3) \cong \mathbb{RP}^3$, with deck transformation $U \mapsto -U$.

2 Lie Algebras of $\mathrm{SO}(3)$ and $\mathrm{SU}(2)$, and Angular Momentum

1. What is a Lie algebra, intuitively? A Lie group G is a group that is also a smooth manifold. Near the identity $e \in G$, the group "looks like" a vector space — the **tangent space** T_eG. This vector space carries a natural bilinear bracket $[\cdot,\cdot]$ inherited from the group multiplication, making it a **Lie algebra** $\mathfrak{g} = \mathrm{Lie}(G)$.

The fundamental idea: instead of studying the curved, global object G, we study the flat, linearised object $\mathfrak{g}$ at the identity. The exponential map

$$\exp : \mathfrak{g} \longrightarrow G, \qquad X \mapsto e^X = \sum_{n=0}^{\infty} \frac{X^n}{n!}$$

connects the two: every one-parameter subgroup of G arises this way.

2. The Lie algebra $\mathfrak{so}(3)$

Generators. The curve $t \mapsto \exp(tJ_i) \in \mathrm{SO}(3)$ passes through the identity at $t = 0$. Its velocity there is J_i itself, so the matrices

$$J_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad J_y = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad J_z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

are tangent vectors at e and span $T_e\mathrm{SO}(3) = \mathfrak{so}(3)$.

Concretely, $\exp(tJ_z)$ is rotation by angle t about the z-axis:

$$\exp(tJ_z) = \begin{pmatrix} \cos t & -\sin t & 0 \\ \sin t & \cos t & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Abstract characterisation. $\mathfrak{so}(3)$ is the space of real 3×3 *skew-symmetric* matrices ($A^T = -A$), since differentiating $R(t)^T R(t) = I$ at $t = 0$ gives $\dot{R}(0)^T + \dot{R}(0) = 0$. This space is 3-dimensional, consistent with $\dim \mathrm{SO}(3) = 3$.

Commutation relations. Direct matrix multiplication gives

$$\boxed{[J_x, J_y] = J_z, \qquad [J_y, J_z] = J_x, \qquad [J_z, J_x] = J_y.}$$

These are the *defining relations* of $\mathfrak{so}(3)$ (equivalently, $[J_i, J_j] = \varepsilon_{ijk} J_k$ using the Levi-Civita symbol).

The bracket $[X, Y] = XY - YX$ is not just a curiosity — it encodes the *non-commutativity* of rotations in $\mathbb{R}^3$.

3. The Lie algebra $\mathfrak{su}(2)$ SU(2) consists of unitary 2×2 matrices with $\det = 1$. Differentiating $U(t)^\dagger U(t) = I$ and $\det U(t) = 1$ at $t = 0$ shows that

$$\mathfrak{su}(2) = \{\, X \in \mathrm{Mat}_{2\times 2}(\mathbb{C}) \mid X^\dagger = -X,\ \mathrm{tr}(X) = 0 \,\},$$

i.e. traceless skew-Hermitian 2×2 matrices. The standard basis is provided by the **Pauli matrices**

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

so that $T_e\mathrm{SU}(2) = \mathfrak{su}(2)$ is spanned by $\{i\sigma_x,\ i\sigma_y,\ i\sigma_z\}$. These satisfy exactly the same bracket relations:

$$[i\sigma_x,\ i\sigma_y] = i\sigma_z \cdot 2, \quad \text{or after rescaling } e_i = \tfrac{i}{2}\sigma_i : \quad [e_x, e_y] = e_z,\ [e_y, e_z] = e_x,\ [e_z, e_x] = e_y.$$

Key conclusion.

$$\mathfrak{su}(2) \ \cong\ \mathfrak{so}(3) \qquad \text{(as Lie algebras).}$$

The two Lie algebras are *identical*, even though the groups SU(2) and SO(3) differ globally (one is simply connected, the other is not). This is the infinitesimal reflection of the double cover $\mathrm{SU}(2) \twoheadrightarrow \mathrm{SO}(3)$.

4. Vector fields generating rotations On $\mathbb{R}^3$, the infinitesimal rotation about the z-axis acts on coordinates (x, y, z) by J_z, giving the vector field

$$\xi_z = -y\,\partial_x + x\,\partial_y,$$

and similarly

$$\xi_x = -z\,\partial_y + y\,\partial_z, \qquad \xi_y = -x\,\partial_z + z\,\partial_x.$$

The Lie bracket of vector fields reproduces: $[\xi_x, \xi_y] = \xi_z$, etc. This is how the Lie algebra acts geometrically on the manifold.

5. Connection to quantum mechanics In quantum mechanics, the **angular momentum operators** are defined (with $\hbar = 1$) by

$$L_i = -i\,\xi_i, \qquad \text{e.g.,} \quad L_z = -i\big(-y\partial_x + x\partial_y\big)\,.$$

Because $[\xi_i, \xi_j] = \varepsilon_{ijk}\xi_k$ and $L_i = -i\xi_i$, we get

$$\boxed{[L_x, L_y] = iL_z, \qquad [L_y, L_z] = iL_x, \qquad [L_z, L_x] = iL_y,}$$

or compactly $[L_i, L_j] = i\varepsilon_{ijk}L_k$. These are precisely the canonical commutation relations of angular momentum. The factor of $-i$ converts skew-Hermitian generators into *Hermitian* (hence observable) operators.

Integer vs. half-integer spin. Because $\mathfrak{su}(2) \cong \mathfrak{so}(3)$, both groups share the same Lie algebra and hence the same *representation theory at the infinitesimal level*. Representations of the Lie algebra $\mathfrak{su}(2)$ are labelled by $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$:

j	dim	lifts to group
0	1	SO(3) and SU(2)
$\frac{1}{2}$	2	SU(2) only (spinor!)
1	3	SO(3) and SU(2)
$\frac{3}{2}$	4	SU(2) only

Half-integer spin representations ($j \in \frac{1}{2}\mathbb{Z}_{\text{odd}}$) do *not* descend to well-defined representations of SO(3): a 2π rotation gives -1, not $+1$. They are honest representations of SU(2), the double cover. **This is precisely why electrons (spin-$\frac{1}{2}$) require** SU(2) **rather than** SO(3)**.**

Summary.

Geometry		Physics
$\mathfrak{so}(3) \cong \mathfrak{su}(2)$	$\longleftrightarrow$	$[L_i, L_j] = i\varepsilon_{ijk}L_k$
$\exp(tJ_i) \in \mathrm{SO}(3)$	$\longleftrightarrow$	$e^{-i\theta\hat{n}\cdot\mathbf{L}}$ (rotation operator)
double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$	$\longleftrightarrow$	integer vs. half-integer spin

The entire theory of angular momentum in quantum mechanics is, at its heart, the **representation theory of the Lie algebra** $\mathfrak{su}(2) \cong \mathfrak{so}(3)$.